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Heterostructure Device Simulation

This chapter describes carrier transport boundary conditions for heterointerfaces.

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Thermionic Emission Current

Conventional transport equations cease to be valid at a heterojunction interface, and currents and energy fluxes at the abrupt interface between two materials are better defined by the interface condition at the heterojunction. In defining thermionic current and thermionic energy flux, Sentaurus Device follows the literature [1].

Using Thermionic Emission Current

To activate the thermionic current model for electrons at a region-interface (material-interface) heterojunction, the keyword `eThermionic` must be specified in the appropriate region-interface (material-interface) `Physics` section. For example:

```
Physics(MaterialInterface="GaAs/AlGaAs") {
   eThermionic
}
```

Similarly, to activate thermionic current for holes, the keyword `hThermionic` must be specified. The keyword `Thermionic` activates the thermionic emission model for both electrons and holes. If any of these keywords is specified in the `Physics` section for a region `Region.0`, where `Region.0` is a semiconductor, the appropriate model will be applied to each `Region.0`–semiconductor interface.

For small particle and energy fluxes across the interface, the condition of continuous quasi-Fermi level and carrier temperature is sometimes used. This option is activated by the keyword `Heterointerface` in the appropriate `Physics` section. In realistic transistors, such an approach might lead to unsatisfactory results [2].

You can change the coefficients of the thermionic emission model in the `ThermionicEmission`parameter set in an interface-specific section in the parameter file:

```
RegionInterface = "regionA/regionB" {
   ThermionicEmission {
      A = 2, 2     # [1]
      B = 4, 4     # [1]
      C = 1, 1     # [1]
   }
}
```

Thermionic Emission Model

Assume that at the heterointerface between material 1 and material 2, the conduction edge jump is positive, that is $\Delta E_C > 0$, where $\Delta E_C = E_{C,2} - E_{C,1}$ (that is, $\chi_1 > \chi_2$). If $J_{n,2}$ and $S_{n,2}$ are the electron current density and the electron energy flux density entering material 2, and $J_{n,1}$ and $S_{n,1}$ are the electron current density and the electron energy flux density leaving material 1, then the interface condition can be written as:

Equation 922

$$J_{n,2} = J_{n,1}$$

Equation 923

$$J_{n,2} = a_n q \left[v_{n,2} n_2 - \frac{m_{n,2}}{m_{n,1}} v_{n,1} n_1 \exp\left(-\frac{\Delta E_C}{kT_{n,1}} \right) \right]$$

Equation 924

$$S_{n,2} = S_{n,1} + \frac{c_n}{q} J_{n,2} \Delta E_C$$

Equation 925

$$S_{n,2} = -b_n \left[v_{n,2} n_2 kT_{n,2} - \frac{m_{n,2}}{m_{n,1}} v_{n,1} n_1 kT_{n,1} \exp\left(-\frac{\Delta E_C}{kT_{n,1}} \right) \right]$$

where the *emission velocities* are defined as:

Equation 926

$$v_{n,i} = \sqrt{\frac{kT_{n,i}}{2\pi m_{n,i}}}$$

and by default, the coefficients in the above equations are $a_n = 2$, $b_n = 4$, and $c_n = 1$, which corresponds to the literature [1]. Similar equations for the hole thermionic current and hole thermionic energy flux are presented here:

Equation 927

$$J_{p,2} = J_{p,1}$$

Equation 928

$$J_{p,2} = -a_p q\left[v_{p,2}p_2 - \frac{m_{p,2}}{m_{p,1}}v_{p,1}p_1 \exp\left(\frac{\Delta E_V}{kT_{p,1}}\right)\right]$$

Equation 929

$$S_{p,2} = S_{p,1} + \frac{c_p}{q}J_{p,2}\Delta E_V$$

Equation 930

$$S_{p,2} = -b_p\left[v_{p,2}p_2 kT_{p,2} - \frac{m_{p,2}}{m_{p,1}}v_{p,1}n_1 kT_{p,1}\exp\left(\frac{\Delta E_V}{kT_{p,1}}\right)\right]$$

Equation 931

$$v_{p,i} = \sqrt{\frac{kT_{p,i}}{2\pi m_{p,i}}}$$

An equivalent set of equations are used if Fermi carrier statistics is selected.

Thermionic Emission Model With Fermi Statistics

With Fermi statistics, the electron current density and the electron energy flux density are:

Equation 932

$$J_{n,2} = a_n q\left(v_{n,2}N_{C,2}\zeta_{n,2} - \frac{m_{n,2}}{m_{n,1}}v_{n,1}N_{C,1}\zeta_{n,1}\right)$$

Equation 933

$$S_{n,2} = -b_n\left(v_{n,2}kT_{n,2}N_{C,2}\zeta_{n,2} - \frac{m_{n,2}}{m_{n,1}}v_{n,1}kT_{n,1}N_{C,1}\zeta_{n,1}\right)$$

Equation 934

$$\zeta_{n,2} = \ln\left[1 + \exp\left(-\eta_{n,2}\right)\right] + \eta_{n,2}$$

Equation 935

$$\zeta_{n,1} = \ln\left[1 + \exp\left(-\eta'_{n,1}\right)\right] + \eta'_{n,1}$$

Equation 936

$$\eta'_{n,1} = \eta_{n,1} - \frac{\Delta E_C}{kT_{n,1}}$$

Here, η_n is given in Equation 50 and N_C is given in Equation 182. Similar equations apply to holes:

Equation 937

$$J_{p,2} = -a_p q\left(v_{p,2}N_{V,2}\zeta_{p,2} - \frac{m_{p,2}}{m_{p,1}}v_{p,1}N_{V,1}\zeta_{p,1}\right)$$

Equation 938

$$S_{p,2} = -b_p\left(v_{p,2}kT_{p,2}N_{V,2}\zeta_{p,2} - \frac{m_{p,2}}{m_{p,1}}v_{p,1}kT_{p,1}N_{V,1}\zeta_{p,1}\right)$$

Equation 939

$$\zeta_{p,2} = \ln\left[1 + \exp\left(-\eta_{p,2}\right)\right] + \eta_{p,2}$$

Equation 940

$$\zeta_{p,1} = \ln\left[1 + \exp\left(-\eta'_{p,1}\right)\right] + \eta'_{p,1}$$

Equation 941

$$\eta'_{p,1} = \eta_{p,1} + \frac{\Delta E_V}{kT_{p,1}}$$

Here, η_p is given in Equation 51 and N_V is given in Equation 186.

If Fermi statistics is used, this model can be activated by specifying `Formula=1` in the `ThermionicEmission` section of the parameter file:

```
ThermionicEmission { Formula=1 }
```

By default `Formula=0`, it activates the old model, where the Boltzmann-like thermionic emission equations similar to Equation 922–Equation 931 are used. This can lead to incorrect results in high carrier density. In this case, a warning message will be given.

Note:

Always specify `Formula=1` when Fermi statistics is used. If Fermi statistics is not used, Equation 922–Equation 931 will be activated regardless of the `Formula` statement in the parameter file.

Gaussian Transport Across Organic Heterointerfaces

A thermionic-like current boundary condition has been introduced to correctly account for carrier transport across organic heterointerfaces. An organic heterointerface is defined in this context as an heterointerface with the Gaussian density-of-states (DOS) model (see Gaussian Density-of-States for Organic Semiconductors on page 358) activated in both regions that are adjacent to the heterointerface.

Using Gaussian Transport at Organic Heterointerfaces

The model is activated by switching to the Gaussian DOS model in both regions of the heterointerface that are meant to be organic:

```
Physics(Region="OrganicRegion_1") {
   EffectiveMass(GaussianDOS)
}
Physics(Region="OrganicRegion_2") {
   EffectiveMass(GaussianDOS)
}
```

and then specifying the keyword `Organic_Gaussian` as an option for the `Thermionic` model in the `Physics` section of the organic heterointerface:

```
Physics(RegionInterface="OrganicRegion_1/OrganicRegion_2") {
   Thermionic(Organic_Gaussian)
}
```

This syntax also automatically switches on the double points at the organic heterointerface.

You can adjust the parameters $v_{n,\mathrm{org}}$ and $v_{p,\mathrm{org}}$ in Equation 943 and Equation 945 in the `ThermionicEmission` section of the parameter file (their default values are 1×10^6 cm/s):

```
RegionInterface="OrganicRegion_1/OrganicRegion_2" {
   ThermionicEmission {
      vel_org = 1e7, 1e7   # [cm/s]
   }
}
```

Gaussian Transport at Organic Heterointerface Model

Assuming a positive conduction edge jump and a negative valence edge jump from material 1 to material 2, the boundary conditions at the organic heterointerface are given by:

Equation 942

$$J_{n,2} = J_{n,1}$$

Equation 943

$$J_{n,2} = v_{n,\mathrm{org}} q \left(n_2 - n_1 \exp\left(-\frac{\Delta E_\mathrm{C}}{kT_{n,1}} \right) \right)$$

Equation 944

$$J_{p,2} = J_{p,1}$$

Equation 945

$$J_{p,2} = v_{p,\mathrm{org}} q \left(p_2 - p_1 \exp\left(-\frac{\Delta E_\mathrm{V}}{kT_{p,1}} \right) \right)$$

where:

- $J_{n,2}$ and $J_{n,1}$ are the electron current densities entering material 2 and leaving material 1, respectively.
- $J_{p,2}$ and $J_{p,1}$ are the hole current densities leaving material 2 and entering material 1, respectively.
- $\Delta E_\mathrm{C} = d_{\mathrm{C},2} - d_{\mathrm{C},1}$ where $d_{\mathrm{C},2}$ and $d_{\mathrm{C},1}$ are the distances of the Gaussian distribution peaks to the conduction band edges.
- $\Delta E_\mathrm{V} = d_{\mathrm{V},2} - d_{\mathrm{V},1}$ where $d_{\mathrm{V},2}$ and $d_{\mathrm{V},1}$ are the distances of the Gaussian distribution peaks to the valence band edges.

References

[1] D. Schroeder, *Modelling of Interface Carrier Transport for Device Simulation*, Wien: Springer, 1994.

[2] K. Horio and H. Yanai, “Numerical Modeling of Heterojunctions Including the Thermionic Emission Mechanism at the Heterojunction Interface,” *IEEE Transactions on Electron Devices*, vol. 37, no. 4, pp. 1093–1098, 1990.